

K-FIELD

Apparent-Weight Modulation Spin-Reversal Experiment

Immediate full-envelope demonstration by reversing tubular-rotor axial spin at a fixed laboratory orientation, with sidereal geometry, scale reaction, odd-channel metrology, and direct force observables

K-Field apparent-weight modulation by controlled spin reversal

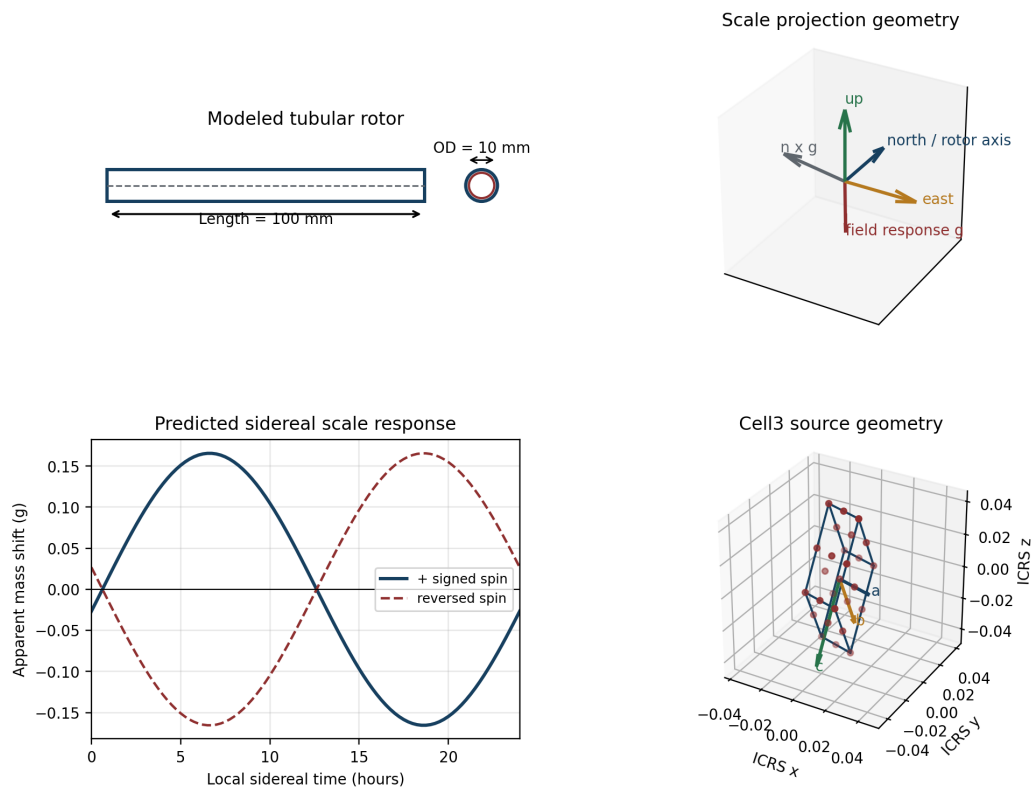


Figure 1. Title composite. The tube dimensions, Cell3 source geometry, local scale projection, and opposite-spin sidereal waveforms are generated from the supplied numerical ledger and the exact fixed-laboratory transformation.

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Frontispiece: authoritative Cell3 base cell, face planes, and all nodes

Authoritative Cell3 base cell: six faces, 12 edges, all 27 nodes

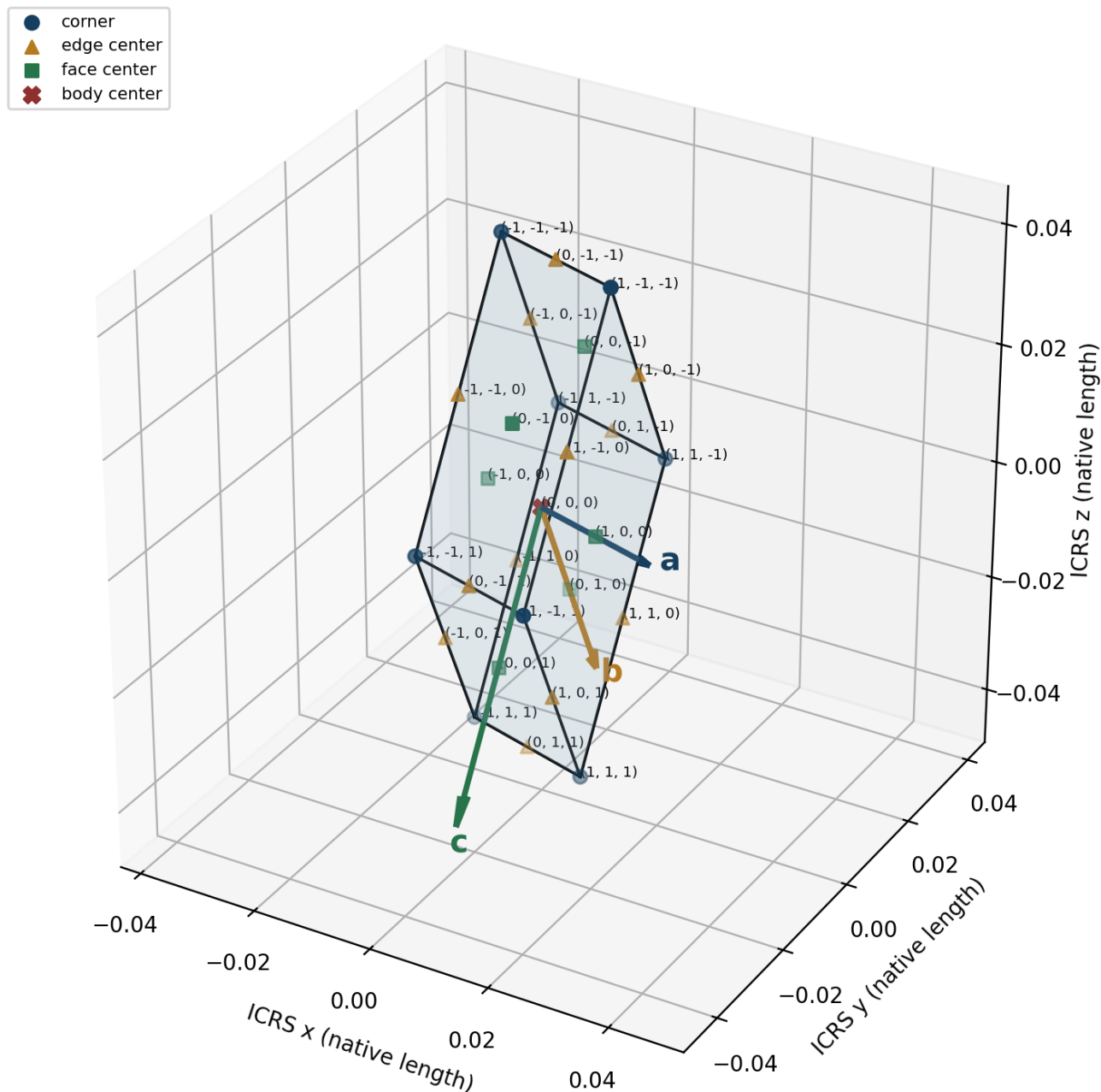


Figure 2. Authoritative Cell3 base cell in ICRS/J2000 Cartesian coordinates. The centered primitive cell is reconstructed from the source basis matrix. All six boundary faces, 12 edges, 27 half-step address nodes, integer node labels, and basis vectors are shown in native dimensions.

Cell3 face-plane system

Cell3 central face planes and source unit normals

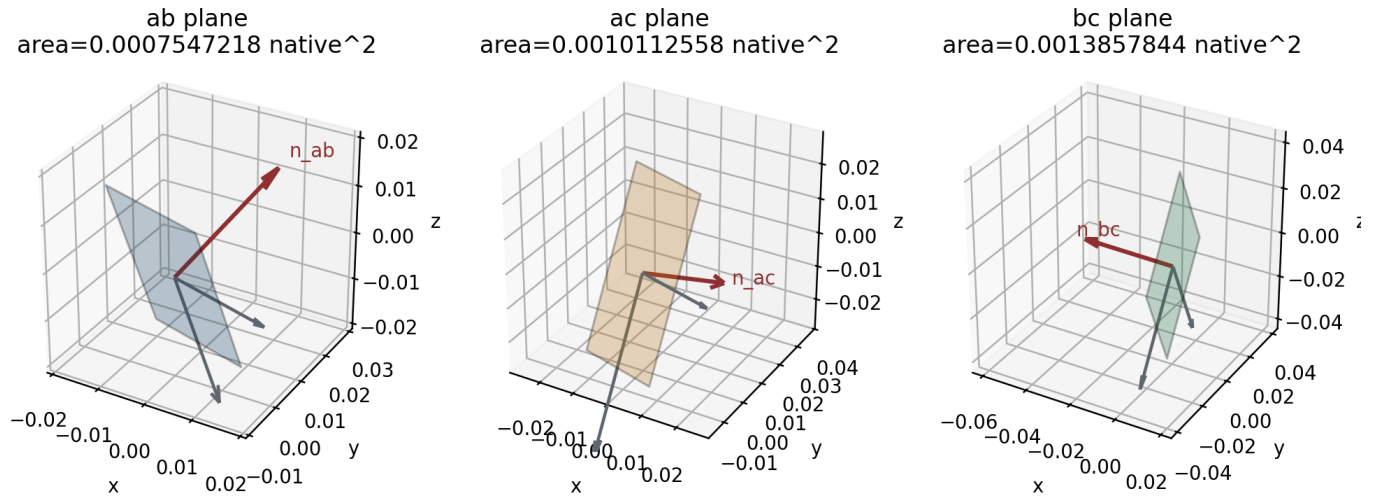


Figure 3. Central ab, ac, and bc planes. Each panel uses the exact basis vectors, native face area, and source unit normal. The plotted coordinates are native ICRS lengths.

The Cell3 geometry is not used as a decorative frame. It supplies the source response matrix and the fixed ICRS orientation from which the laboratory force law is evaluated. The base-cell dimensions and face normals therefore remain part of the complete measurement definition.

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This paper is a standalone archival and educational reference. It begins with the complete integrated tubular-rotor force law and derives the response of a weigh scale at a fixed Earth laboratory. It defines the ICRS source vectors, the rotating local east-north-up basis, the scale normal-force equation, the globally optimal passive rotor orientation, the exact sidereal waveform, numerical amplitudes for the supplied 10 mm by 100 mm stainless-steel tube at 50,000 rpm, spin-reversal observables, multi-rotor scaling, experimental controls, reconstruction equations, and complete numerical ledgers. All quantitative figures are generated from the supplied machine-readable source and the equations stated in this report.

Abstract

A complete thin-walled tube rotating about its longitudinal axis produces the integrated force $F_K = (\kappa_F m_r \omega / 2) [n \times g]$, where $g = G_{\text{phase}} V_E$ is fixed in the ICRS frame over the short daily experiment and n is the signed rotor axis. A laboratory scale measures only the force component along the local vertical. Because the local vertical and horizontal basis rotate with Earth, a rotor held at a fixed laboratory orientation produces a sidereal modulation even though its bearings and scale remain stationary on the floor. The exact scale law is $W_{\text{scale}} = M_{\text{assembly}} g_0 - F_K \cdot U$. For a horizontal north-south rotor, $U \times N = -E$, so the measured force becomes $F_U = -C E \cdot g$, with $C = \kappa_F m_r \omega / 2$. The east unit vector rotates entirely in the ICRS equatorial plane and therefore samples the full equatorial component of g once per sidereal day. This makes the horizontal north-south axis the globally maximal passive orientation for daily modulation at every nonpolar laboratory latitude.

For the supplied rotor, $m_r = 0.022619467106$ kg, $\omega = 5235.987755982988$ rad/s at 50,000 rpm, and the source maximum total force is $1.694455587915e-03$ N. The equatorial fraction of the response is $g_{\text{perp}}/|g| = 0.958191446445$. The resulting north-south vertical-force amplitude is $1.623612850722e-03$ N, equivalent to 0.165562 g of apparent mass. A single run therefore has a mean-to-peak amplitude of 0.1656 g and a peak-to-peak range of 0.3311 g. Reversing spin reverses the complete waveform. The difference $W(+\omega) - W(-\omega)$ has amplitude 0.3311 g and peak-to-peak range 0.6622 g while rejecting spin-even offsets.

Central result

A fixed laboratory does not remove the directional force from measurement. Earth rotation continuously changes the projection of the ICRS-locked force onto the local gravity axis. With the rotor horizontal and aligned true north-south, the predicted scale signal is a phase-locked sidereal sinusoid whose amplitude is 0.1656 g for one modeled tube at 50,000 rpm.

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1. Scope and measurement objective

The apparatus is a complete tubular rotor, motor, bearing support, and enclosure placed on a weigh scale at a fixed Earth laboratory. The rotor spins about its own longitudinal axis while the complete assembly remains stationary relative to the floor. The laboratory is not stationary in the ICRS frame: Earth rotation carries the local east-north-up triad through a full orientation cycle once per sidereal day, while Earth translation supplies the velocity vector that enters the field response.

The measurement objective is the normal force applied to the scale. The scale does not directly report the complete three-dimensional K-Field force. It reports the mechanical support force required along the scale normal, which is chosen to coincide with the local upward direction $\mathbf{U}$. Any horizontal K-Field component is reacted by bearings, feet, or lateral constraints and does not enter the ideal scale channel except through mechanical cross-coupling.

The experimental strategy is therefore to choose a rotor axis for which the vertical projection changes as strongly as possible during Earth rotation. The result is a passive orientation: the rotor axis remains fixed in the laboratory, while the rotating Earth performs the orientation sweep.

Measurement quantities

Quantity	Definition	Units
M_{assembly}	Total mass supported by scale	kg
m_r	Active tubular rotor mass	kg
$\mathbf{U}$	Local upward unit vector	unitless
$\mathbf{n}$	Signed rotor spin-axis unit vector	unitless
ω	Signed axial angular speed	rad/s
$\mathbf{g}$	$G_{\text{phase}} \mathbf{V}_E$ field-response vector	m/s effective response
F_K	Integrated force on active rotor	N
W_{scale}	Scale normal force	N
Δm_{scale}	Equivalent scale mass shift	kg or g

2. Source geometry and numerical lock

The force computation is locked to the ICRS/J2000 Cartesian frame. The source Cell3 basis is an oblique three-vector basis with lengths and face areas listed below. The source response matrix acts on the Earth velocity vector to form the field-response vector g . No terrestrial coordinate transformation is applied until the laboratory projection stage.

Quantity	Value	Role
a vector	(0.020381438516946, -0.003416414496411, -0.002332965916729)	native ICRS basis
b vector	(0.002198745968323, 0.012902170258693, -0.034066501536056)	native ICRS basis
c vector	(-0.000630379485347, -0.024257722017128, -0.042924354571094)	native ICRS basis
a	0.020797058782433	native length
b	0.036494205130727	native length
c	0.049308565899859	native length
primitive volume	2.842066937809649e-05	native ³

Plane	Area (native ²)	Included angle (deg)	Unit normal in ICRS
ab	7.547217896726240e-04	83.933483012313	(0.194092200808, 0.913177168821, 0.358379234791)
ac	1.011255757688065e-03	80.448130018338	(0.089052594102, 0.866576769291, -0.491033948326)
bc	1.385784431733287e-03	50.363231813756	(-0.995965190309, 0.083602162507, -0.032619290564)

Cell3 central face planes and source unit normals

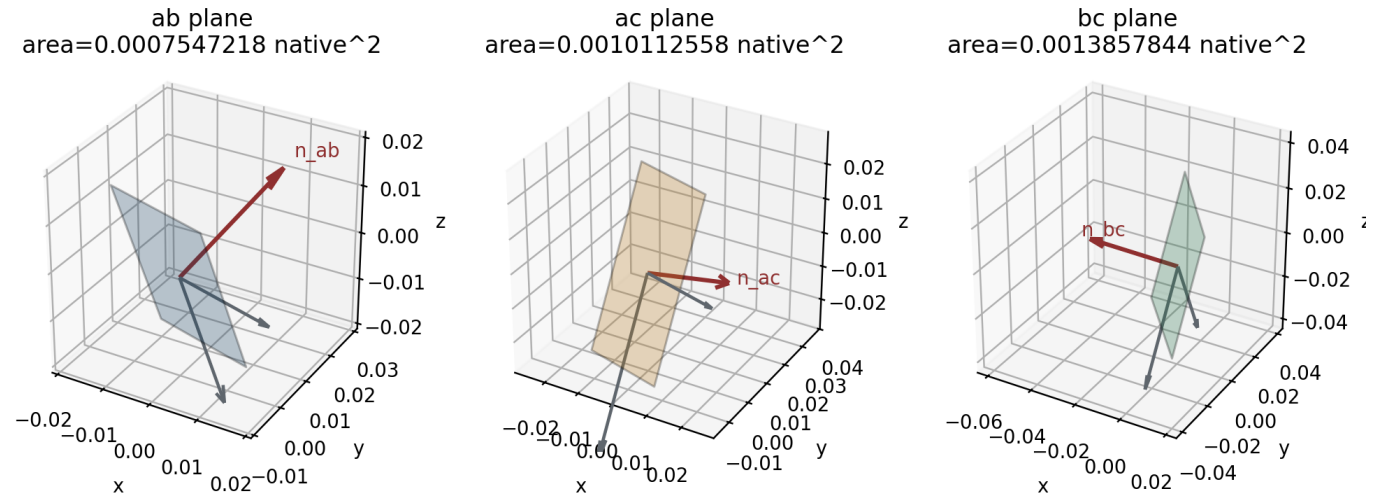


Figure 4. The three central face planes, with exact source areas and normals, provide the dimensional geometric reference for the response construction.

Field and coupling inputs

Quantity	Value	Unit or role
Earth CMB unit vector	(-0.970770855836, 0.207348009111, -0.120874929482)	unitless ICRS
Earth CMB RA, Dec	167.943299880 deg, -6.942599922 deg	direction
Earth velocity V_E	(-359003.199066, 76679.886034, -44701.060101)	m/s ICRS
Field response g	(-880221.981635, -143911.436562, 266335.026949)	m/s effective response
g	930825.217270434834	m/s effective response
g equatorial magnitude	891908.761324178427	m/s effective response
g RA, Dec	189.285391176 deg, 16.626277960 deg	ICRS direction
kappa_F	3.0740508997020345e-11	calibrated coupling
kappa_F interval	[3.0427399447546500e-11, 3.1053618546494197e-11]	coupling interval

The daily scale amplitude depends on the equatorial magnitude $\sqrt{g_x^2 + g_y^2}$, not on the full magnitude $|g|$, because the east unit vector is confined to the ICRS equatorial plane. The axial component g_z contributes to a constant offset for some east-west orientations but does not increase the north-south sidereal harmonic.

3. Integrated tubular-rotor force law

A thin-walled tube contains all circumferential phases at every instant. Integrating the elemental response around the complete wall removes a rotating point-force pulse and leaves a steady center-of-mass force for fixed n , ω , and g . The complete-tube law is linear in active rotor mass and signed angular speed.

$$\begin{aligned} g &= G_{\text{phase}} V_E \\ C &= \kappa_F m_r \omega / 2 \\ F_K &= C (n \times g) \\ a_K &= F_K / m_r = (\kappa_F \omega / 2) (n \times g) \end{aligned}$$

The sign of ω carries the spin direction. Reversing physical spin changes ω to $-\omega$ and reverses every component of F_K without changing its magnitude. Reversing the geometric axis label n while preserving the same physical angular-velocity vector must not be counted as an independent reversal; the signed product ωn is the physical quantity.

The source free-space optimum chooses n perpendicular to g so that $|n \times g| = |g|$. That construction maximizes the total force magnitude. A weigh scale has a different optimization target: it maximizes only $U \cdot F_K$ over the daily Earth-fixed sweep. Consequently, the free-space optimum ICRS axis and the fixed-laboratory optimum passive axis are distinct optimization problems.

Source free-space reference

Quantity	Value
Source optimal tube axis RA, Dec	116.050867377 deg, -44.008760899 deg
Source force direction RA, Dec	84.082114314 deg, 41.289922334 deg
Maximum total force at 50,000 rpm	1.694455587915101e-03 N
Maximum free acceleration	7.491138407399218e-02 m/s ²
Force direction dot Earth-CMB vector	1.4668023673069506e-18

4. What a weigh scale measures

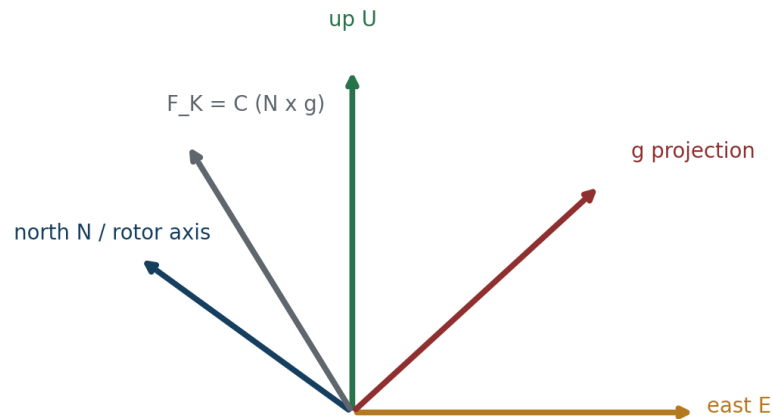
Let the scale platform normal be the local upward unit vector U . The complete assembly has ordinary gravitational force $-M_{\text{assembly}} g_0 U$ and the rotor has the additional K-Field force F_K . Static vertical force balance gives the scale normal force N_{scale} .

$$\begin{aligned} N_{\text{scale}} U + F_K - M_{\text{assembly}} g_0 U &= 0 \\ W_{\text{scale}} = N_{\text{scale}} &= M_{\text{assembly}} g_0 - F_K \cdot U \\ \Delta W = W_{\text{scale}} - M_{\text{assembly}} g_0 &= -F_K \cdot U \\ \Delta m_{\text{scale}} = \Delta W / g_0 &= -(F_K \cdot U) / g_0 \end{aligned}$$

An upward K-Field component reduces the scale reading. A downward component increases it. The baseline assembly mass affects the total displayed weight and the fractional modulation, but it does not change the absolute K-Field force generated by a specified active rotor.

Horizontal force components are not lost. They are carried by bearing reactions and structural supports. They become scale artifacts only when the apparatus has tilt, compliance, friction, cable forces, or load-cell cross-axis sensitivity. A correct design therefore separates the desired vertical projection from horizontal reaction paths.

Fixed-laboratory scale projection for a horizontal north-south rotor



$$\text{Scale channel: } F_U = U \cdot F_K = C (U \times N) \cdot g = -C E \cdot g$$

Only the vertical projection changes the normal force reported by the scale.

Figure 5. Local projection identity for a north-south rotor. The scale channel is the scalar triple product $C (U \times N) \cdot g = -C E \cdot g$.

5. Earth-fixed laboratory basis in ICRS

At geocentric latitude ϕ and local sidereal angle θ , the local unit vectors expressed in ICRS Cartesian coordinates are:

$$\begin{aligned} U(\theta, \phi) &= (\cos(\phi) \cos(\theta), \cos(\phi) \sin(\theta), \sin(\phi)) \\ E(\theta) &= (-\sin(\theta), \cos(\theta), 0) \\ N(\theta, \phi) &= (-\sin(\phi) \cos(\theta), -\sin(\phi) \sin(\theta), \cos(\phi)) \end{aligned}$$

These vectors form a right-handed orthonormal basis: $E \times N = U$, $N \times U = E$, and $U \times E = N$. Earth rotation advances θ through 2π in one sidereal day. The source vector $\mathbf{g}$ remains approximately fixed in ICRS over a single daily run, so the time dependence enters through E , N , and U .

Earth-fixed local basis rotates through the ICRS-locked response vector

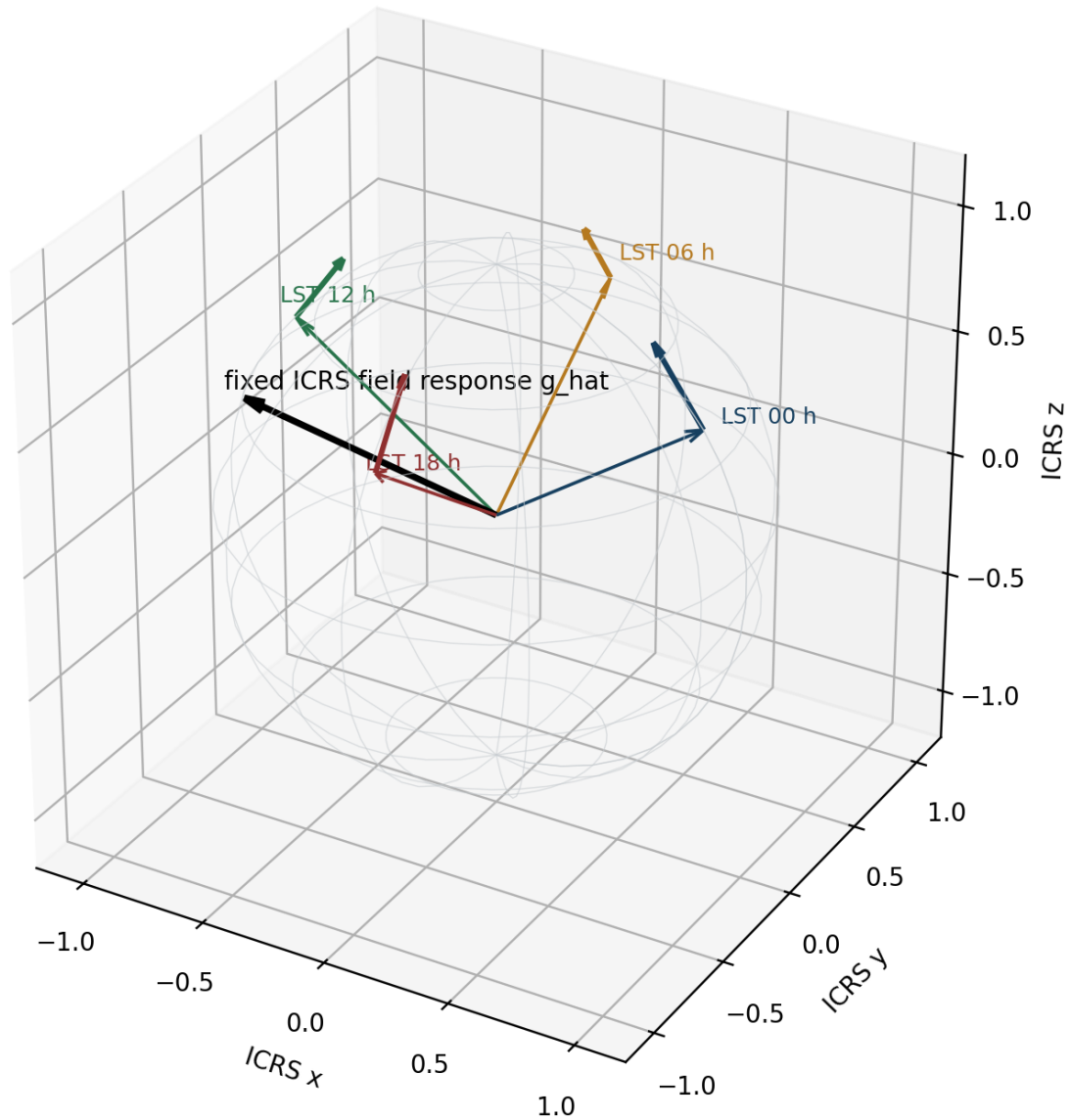


Figure 6. Earth-fixed local basis vectors at four sidereal phases. The displayed 35 degree latitude is a geometric visualization datum; the north-south harmonic-amplitude proof is valid for every latitude.

The daily modulation is therefore an orientation-projection effect. Earth translation is already present in V_E and establishes g . Earth rotation then changes how the stationary laboratory projects the resulting force onto local gravity.

6. General fixed-axis vertical-force law

Write a fixed laboratory rotor axis in local east-north-up components: $n = n_E E + n_N N + n_U U$. The vertical force follows from the scalar triple product.

$$\begin{aligned} F_U &= U \cdot F_K = C U \cdot (n \times g) \\ F_U &= C (U \times n) \cdot g \\ U \times n &= n_E N - n_N E \\ F_U &= C (n_E N - n_N E) \cdot g \end{aligned}$$

The vertical component n_U does not contribute. A vertical rotor axis therefore produces no direct ideal scale modulation because $U \times n = 0$. Any measured vertical-axis signal would be a cross-axis mechanical artifact or a different physical channel, not the direct projection derived here.

For a horizontal axis at azimuth α measured east of true north, $n = \cos(\alpha) N + \sin(\alpha) E$. The vertical force separates into a constant term and a sidereal harmonic:

$$\begin{aligned} F_U(\theta) &= C [\sin(\alpha) N(\theta, \phi) - \cos(\alpha) E(\theta)] \cdot g \\ \text{DC term} &= C \sin(\alpha) \cos(\phi) g_z \\ \text{harmonic amplitude } A(\phi, \alpha) &= C g_{\text{perp}} \sqrt{\cos(\alpha)^2 + \sin(\alpha)^2 \sin(\phi)^2} \end{aligned}$$

The formula is exact for the fixed source vector and ideal local basis. It supplies the complete passive-axis optimization and separates daily modulation from constant offset.

7. Proof of the optimum passive spin axis

For any nonpolar latitude, $|\sin(\phi)|$ is less than one. The harmonic amplitude contains $\cos(\alpha)^2$ with unit coefficient and $\sin(\alpha)^2$ with the smaller coefficient $\sin(\phi)^2$. Therefore the maximum occurs when $\sin(\alpha) = 0$, or $\alpha = 0$ or 180 degrees. These are the two signed directions along the same horizontal north-south axis.

$$A(\phi, \alpha)^2 / (C^2 g_{\text{perp}}^2) = \cos(\alpha)^2 + \sin(\alpha)^2 \sin(\phi)^2$$

$$A_{\text{max}} = C g_{\text{perp}} \text{ at } \alpha = 0 \text{ deg or } 180 \text{ deg}$$

$$n_{\text{opt, lab}} = \pm N$$

At a geographic pole, every horizontal azimuth lies in the ICRS equatorial plane and the azimuthal distinction becomes degenerate. At all other latitudes, the horizontal north-south axis uniquely maximizes the daily harmonic. This maximum is independent of latitude.

A horizontal east-west rotor produces a harmonic reduced by $|\sin(\phi)|$ and also produces a constant vertical offset $C \cos(\phi) g_z$. At the equator, its daily harmonic vanishes and only the constant offset remains. A north-south rotor has zero constant term and the full equatorial harmonic, which is advantageous for drift rejection.

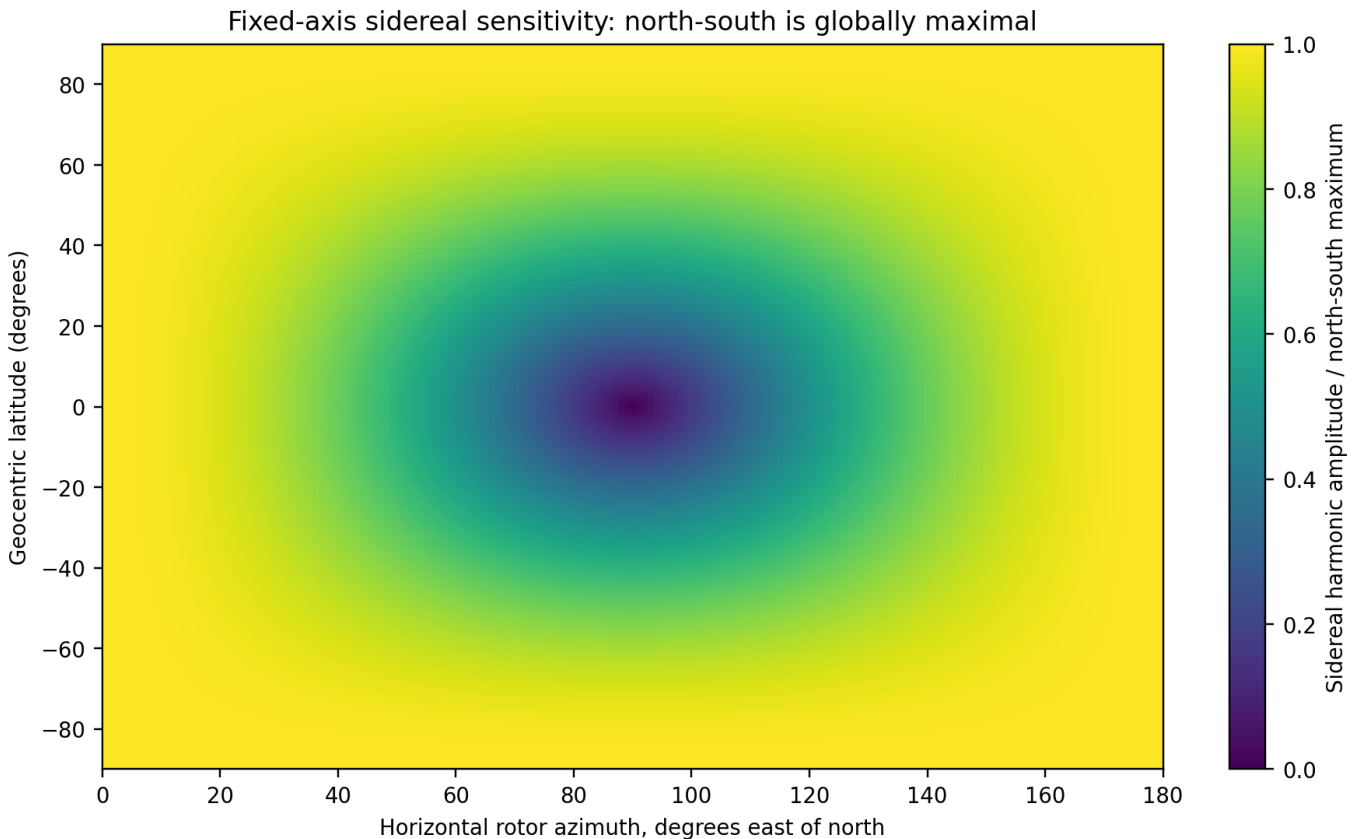


Figure 7. Exact normalized sidereal harmonic amplitude as a function of latitude and horizontal rotor azimuth. North-south is maximal at every nonpolar latitude; east-west sensitivity falls to zero at the equator.

8. North-south rotor waveform

Set $n = N$ for the adopted positive axis. Because $U \times N = -E$, the vertical force is the negative eastward projection of g .

$$\begin{aligned} F_U(\theta) &= -C E(\theta) \cdot g \\ E(\theta) \cdot g &= -g_x \sin(\theta) + g_y \cos(\theta) \\ F_U(\theta) &= C [g_x \sin(\theta) - g_y \cos(\theta)] \\ F_U(\theta) &= C g_{\text{perp}} \sin(\theta - RA_g) \end{aligned}$$

The source field-response right ascension is $RA_g = 189.285391176$ degrees. The waveform is a pure first sidereal harmonic under the fixed-vector daily approximation. Its mean is zero, its amplitude is $C g_{\text{perp}}$, and reversing spin shifts the waveform by 180 degrees in phase.

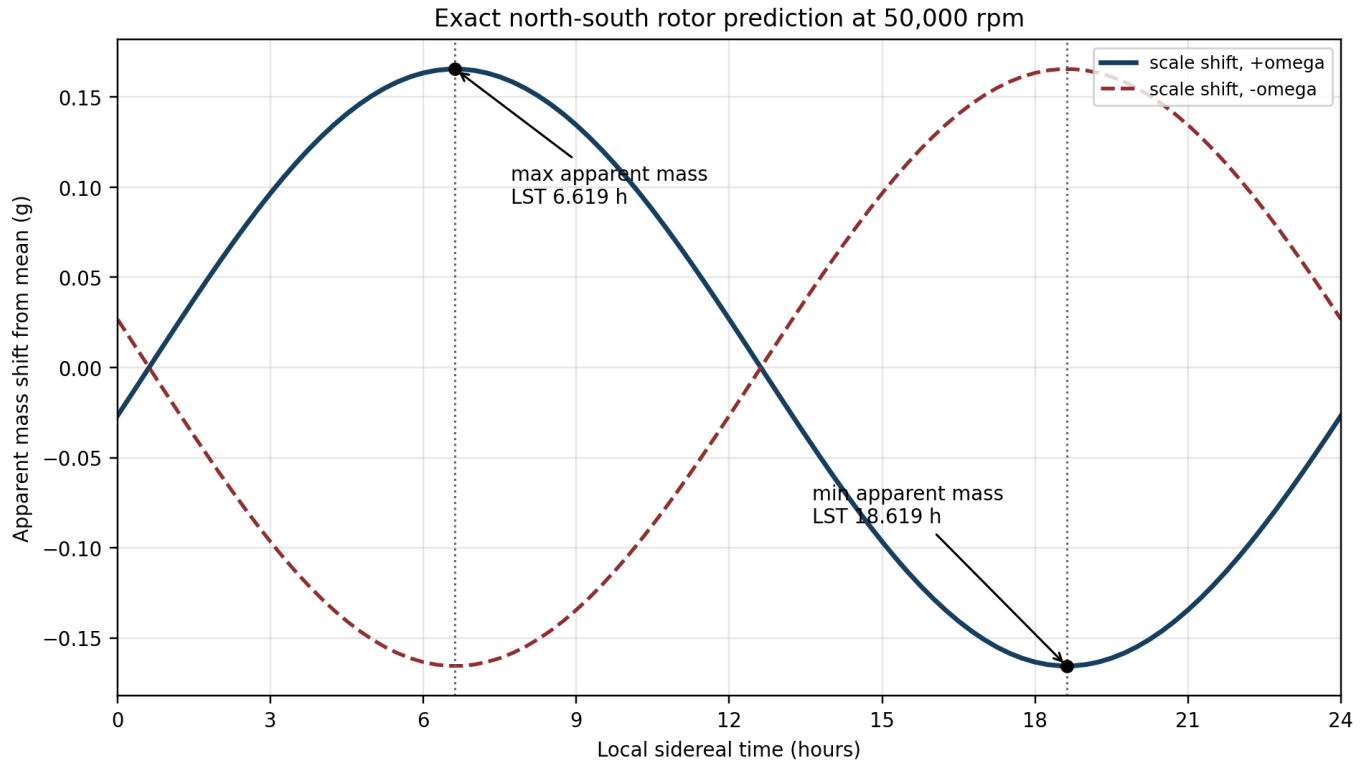


Figure 8. Predicted apparent-mass waveforms for positive and reversed spin. The ordinate is the change from the mean scale reading, expressed as equivalent grams.

9. Numerical prediction for the supplied tube

Rotor property	Value	Unit
Material working density	8000.0	kg/m ³
Outside diameter	0.010000	m
Inside diameter	0.008000	m
Wall thickness	0.001000	m
Length	0.100000	m
Raw tube mass	0.022619467105847	kg
Spin speed	50000.0	rpm
Angular speed	5235.987755982988347	rad/s
Outer surface speed	26.179938779915	m/s
Axial moment of inertia	4.636990756698536e-07	kg m ²
Stored rotational energy	6.356286719461	J

Force and scale conversion

Derived quantity	Value
$C = \kappa_F m_r \omega / 2$	1.820379977332315e-09
Maximum total force $C g $	1.694455587915101e-03 N
Equatorial response fraction $g_{\text{perp}}/ g $	0.958191446445364
North-south vertical-force amplitude	1.623612850721801e-03 N
Equivalent apparent-mass amplitude	0.165562435 g
Single-run peak-to-peak range	0.331124869 g
Amplitude interval from κ_F bounds	[0.163876087, 0.167248782] g

The active tube weighs approximately 0.221821 N conventionally. The predicted north-south K-Field vertical amplitude is 1.623613e-03 N, equal to 0.731947% of the tube conventional weight. The absolute scale shift is set by active rotor mass and spin, not by the mass of the housing.

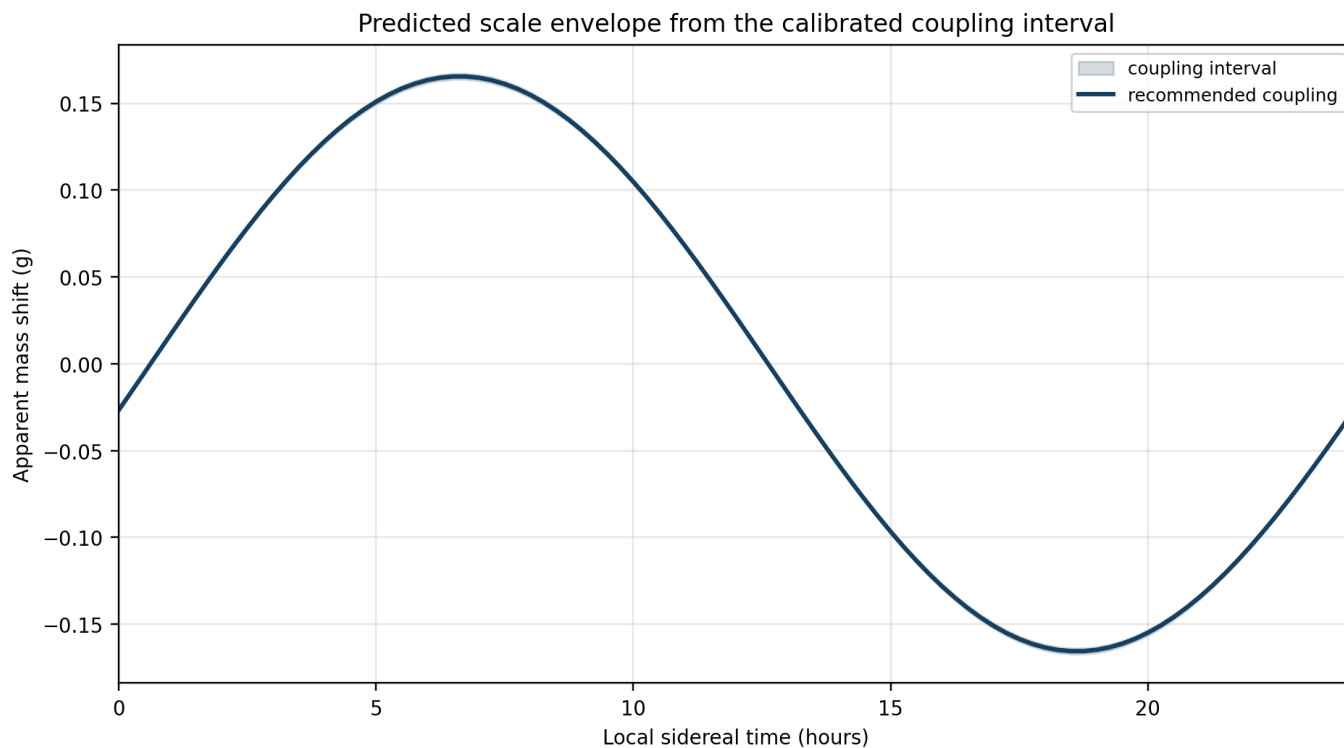


Figure 9. Coupling-interval envelope for one positive-spin daily run. The width reflects only the source κ_F interval.

10. Phase, extrema, and sidereal timing

For positive signed spin about +N, upward K-Field force is maximal when $\theta - RA_g = +90$ degrees. The scale reading is then minimal. Downward force is maximal 12 sidereal hours later and the scale reading is maximal.

Event	Local sidereal angle	Local sidereal time	Scale response
Maximum upward K-Field force	279.285391176 deg	18.619026078 h	minimum apparent mass
Maximum downward K-Field force	99.285391176 deg	6.619026078 h	maximum apparent mass
Zero crossing 1	189.285391176 deg	12.619026078 h	mean scale reading
Zero crossing 2	9.285391176 deg	0.619026078 h	mean scale reading

Clock time depends on longitude and calendar date because local sidereal time advances relative to civil solar time. The fundamental period is the sidereal day, approximately 86,164.0905 seconds. A scale signal locked to sidereal phase rather than solar phase is a central directional signature.

The annual change of Earth velocity alters g slowly and can shift amplitude and phase. For a single-day first experiment, the fixed g vector supplied by the source ledger is the correct reference approximation. Multi-month analysis should recompute $V_E(t)$, $g(t)$, and the predicted local projection for each sample.

11. Immediate full-envelope spin reversal

At a fixed laboratory orientation, reversing the signed axial angular velocity reverses the complete K-Field force vector. The result follows directly from the linear dependence on signed angular speed.

```
omega -> -omega
F_K(+omega) = (kappa_F m_r omega / 2) [n x g]
F_K(-omega) = -F_K(+omega)
```

For a scale, let F_U be positive upward. The two matched steady-spin readings are

```
W_+ = M_assembly g0 - F_U
W_- = M_assembly g0 + F_U
W_- - W_+ = 2 F_U
[W_+ + W_-] / 2 = M_assembly g0
```

The midpoint of the opposite-spin readings is the ordinary nonrotating weight, while their separation is twice the instantaneous vertical K-Field force. At the optimum sidereal phase and optimum fixed rotor-axis orientation, spin reversal immediately traverses the entire vertical force envelope: $+F_{U,max} \leftrightarrow -F_{U,max}$.

For the modeled 100 mm tube at 50,000 rpm, $F_{U,max} = 1.623613$ mN. The two steady spin states therefore differ by 3.247226 mN. Expressed as scale-indicated mass, the full reversal separation is 0.331125 g.

```
M_0 - 0.165562 g -> M_0 -> M_0 + 0.165562 g
+omega steady zero speed -omega steady
```

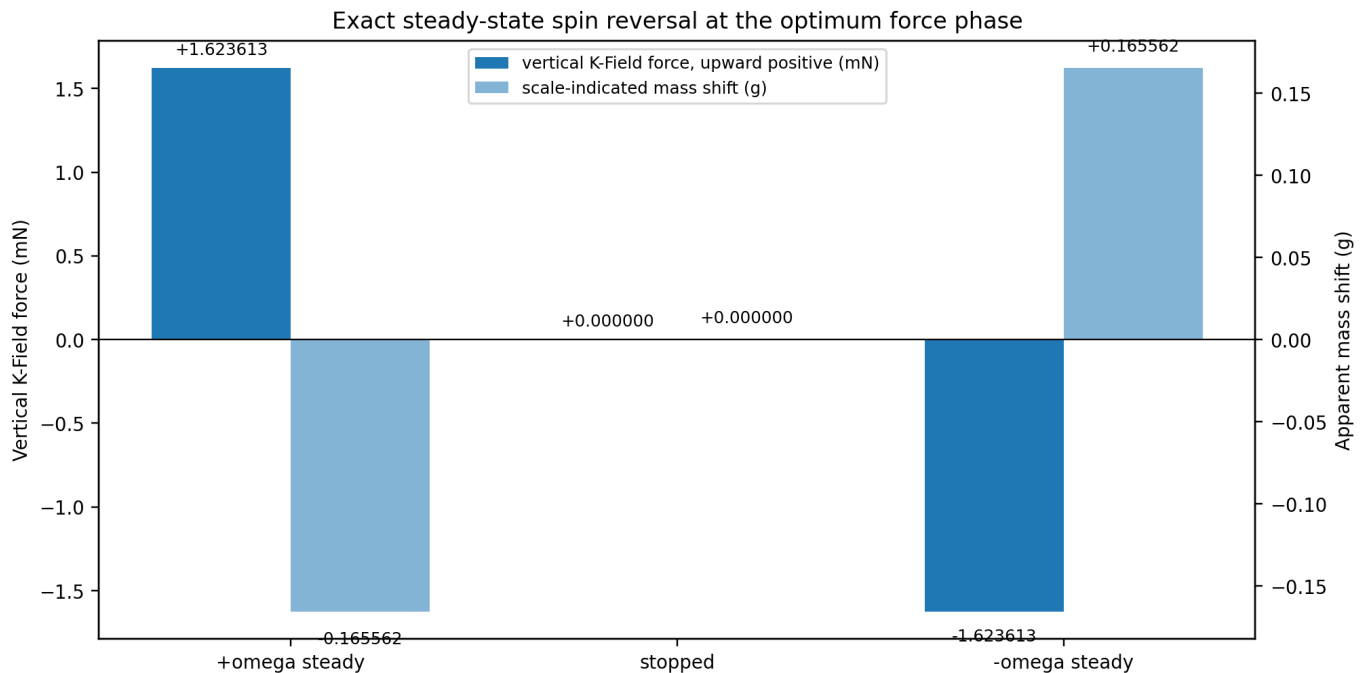


Figure 10. Exact steady-state result at the optimum force phase. Positive and negative matched spin produce equal and opposite vertical force and equal and opposite scale shifts about the stopped midpoint.

This controlled reversal reproduces within one laboratory session the two extrema that Earth rotation would otherwise present approximately 12 sidereal hours apart. The qualification is geometric: reversal produces the full instantaneous envelope $2|F_U(t)|$. It equals the global maximum $2F_{U,max}$ only when the laboratory is near the optimum vertical-projection phase. At other sidereal phases the same reversal remains exactly antisymmetric, but the separation is smaller according to the daily projection.

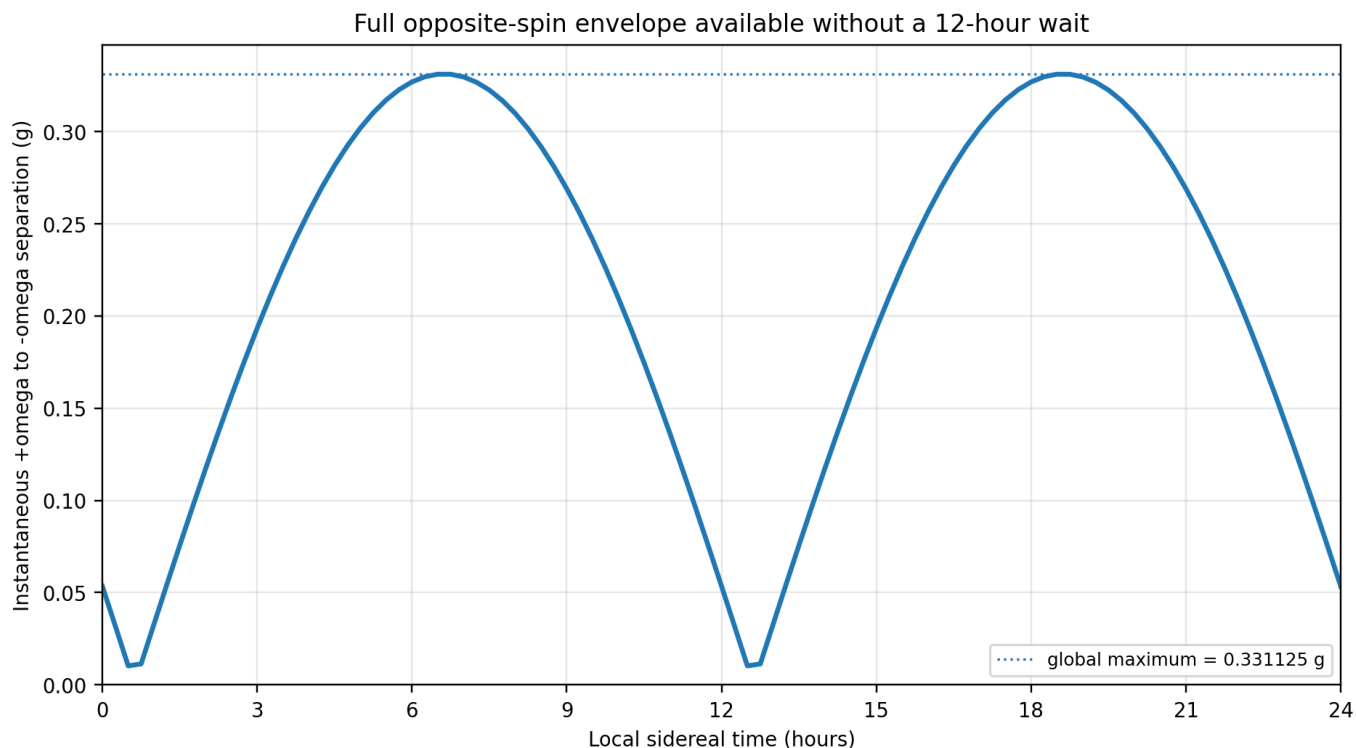


Figure 11. Instantaneous opposite-spin separation as a function of local sidereal time. The curve is 2 times the absolute value of the single-spin scale shift. A reversal at the optimum phase gives the global 0.3311 g separation without waiting for Earth to rotate to the opposite phase.

The reversal also defines a strong synchronous measurement. The half-difference isolates the spin-odd force channel, and the half-sum estimates the spin-even baseline.

$$S_{\text{odd}}(t) = [W_{-\omega}(t) - W_{+\omega}(t)] / 2 = F_U(t)$$

$$S_{\text{even}}(t) = [W_{-\omega}(t) + W_{+\omega}(t)] / 2 = M_{\text{assembly}} g_0 + \text{spin-even background}$$

Alternating between equal-magnitude positive and negative spin states converts the slowly varying sidereal signal into a controlled spin-polarity signal. Thermal scale drift, static imbalance, gravitational tides, housing mass, and most static mechanical loads remain predominantly spin even and cancel in the differenced result. The desired response reverses exactly because ω changes sign while the apparatus orientation and field vector remain unchanged.

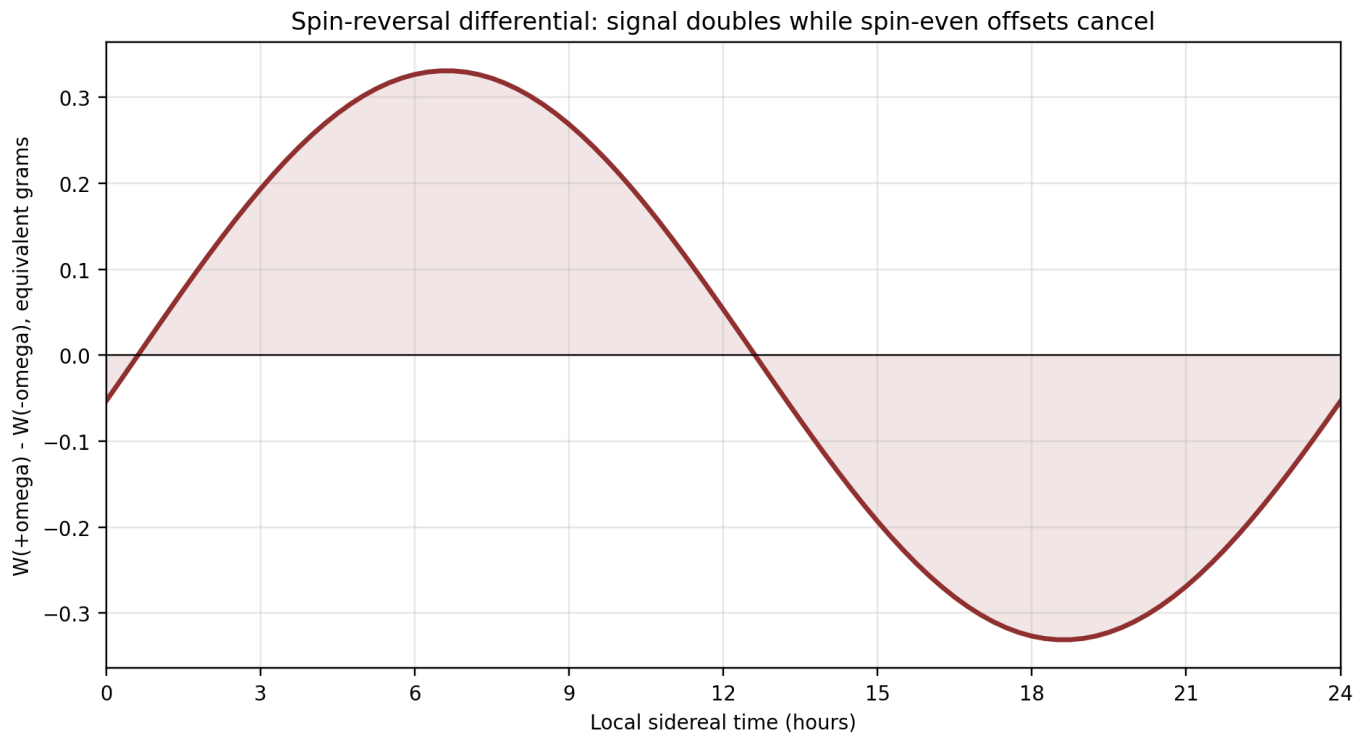


Figure 12. Signed full reversal difference over one sidereal day. The desired spin-odd waveform is twice the single-run shift and reverses through zero at the same sidereal phases.

Reversal data must be evaluated only after each direction reaches the same stable speed magnitude. Samples acquired during deceleration, zero crossing, acceleration, and mechanical settling contain motor torque, bearing-load redistribution, gyroscopic reaction, vibration, and thermal transients. These transition intervals are diagnostic but are not the primary force estimate.

```
+omega_0 steady -> transition excluded -> -omega_0 steady
equal |omega|, matched temperature, matched averaging duration
```

The clean comparison uses identical signed-speed magnitude, rotor position, enclosure state, motor operating condition, scale range, and averaging duration in the two steady blocks. Repeated ABBA or BAAB ordering prevents monotonic scale drift from being mistaken for a spin-odd result.

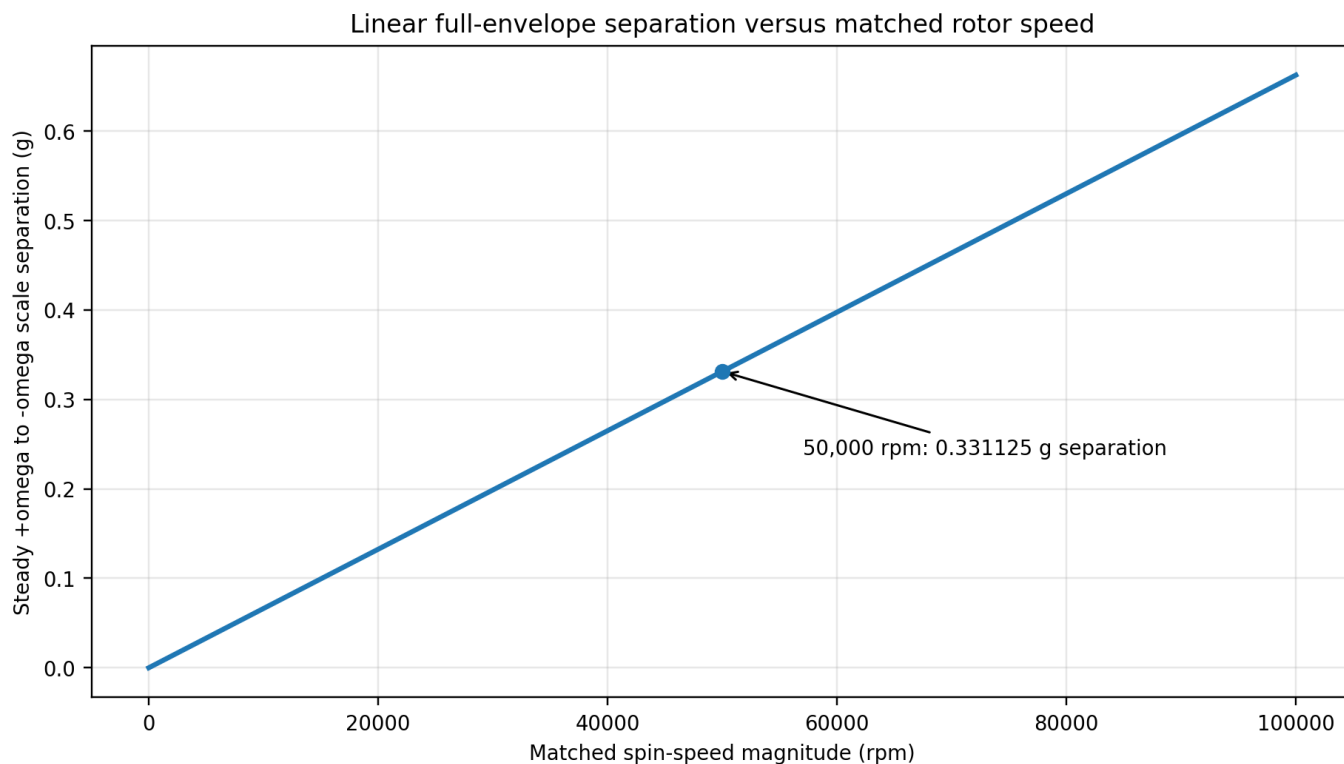


Figure 13. Full steady opposite-spin separation versus matched rotor speed, calculated from the complete source spin-rate ledger. The separation is linear in $|\omega|$.

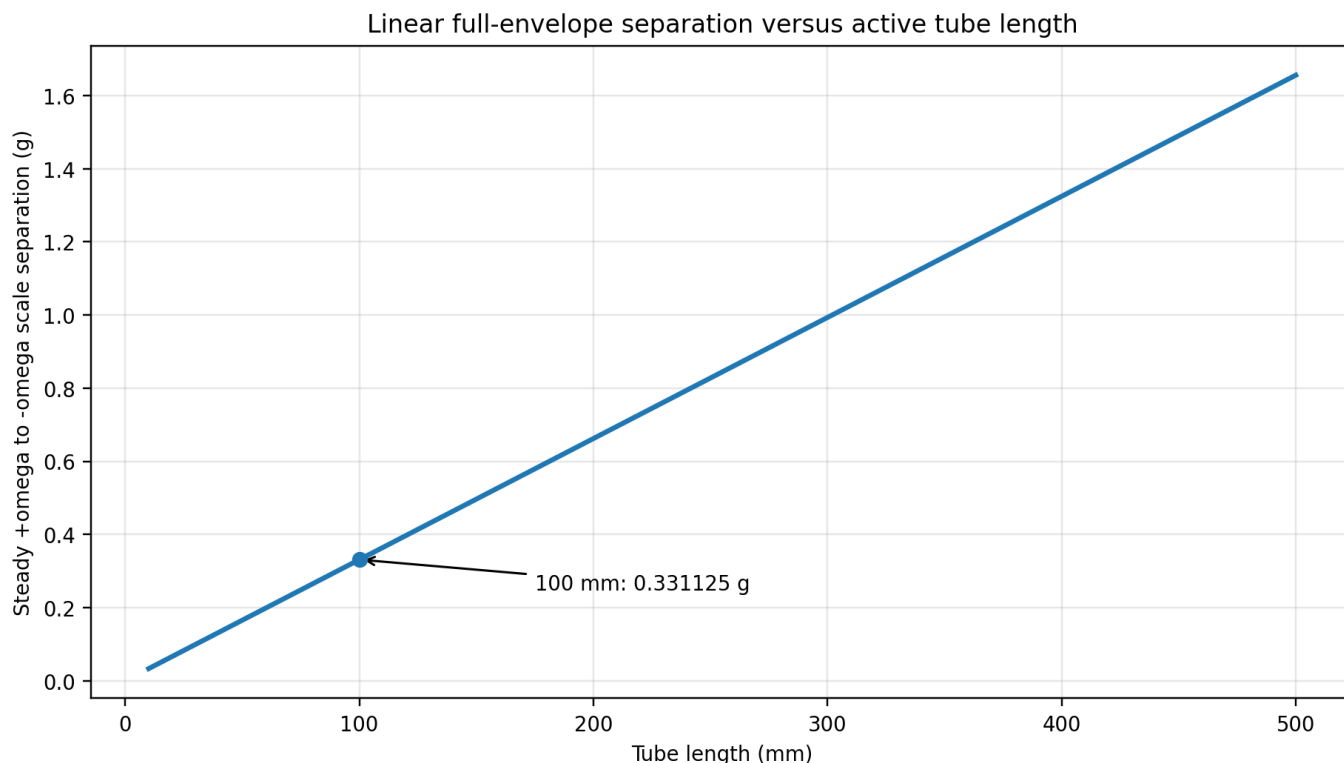


Figure 14. Full steady opposite-spin separation versus active tube length, calculated from the exact source length sweep. For fixed section and speed, active mass and reversal separation are linear in length.

12. Dependence on latitude and azimuth

The north-south harmonic amplitude is independent of latitude. Other horizontal azimuths mix the rotating east vector with the latitude-dependent north vector. The table gives normalized harmonic amplitude and constant offset for representative parameter values, using the exact source g vector.

Latitude	Azimuth	Axis description	Harmonic amplitude (g)	DC offset (g)
0 deg	0 deg	north-south	0.165562	0.000000
0 deg	45 deg	diagonal	0.117070	0.034959
0 deg	90 deg	east-west	0.000000	0.049439
30 deg	0 deg	north-south	0.165562	0.000000
30 deg	45 deg	diagonal	0.130889	0.030275
30 deg	90 deg	east-west	0.082781	0.042815
45 deg	0 deg	north-south	0.165562	0.000000
45 deg	45 deg	diagonal	0.143381	0.024719
45 deg	90 deg	east-west	0.117070	0.034959
60 deg	0 deg	north-south	0.165562	0.000000
60 deg	45 deg	diagonal	0.154869	0.017479
60 deg	90 deg	east-west	0.143381	0.024719
90 deg	0 deg	north-south	0.165562	0.000000
90 deg	45 deg	diagonal	0.165562	0.000000
90 deg	90 deg	east-west	0.165562	0.000000

The DC offset changes sign with spin and can be measured by reversal, but it is more vulnerable to scale tare and long-term drift than the sidereal harmonic. The north-south axis avoids this offset and concentrates the response into a clean periodic channel.

13. Spin-rate, length, and rotor-count scaling

For fixed geometry and orientation, force is linear in active rotor mass and signed angular speed. A tube of the same cross-section has mass proportional to length, so its absolute scale signal is proportional to length. Free acceleration remains independent of length because both force and mass scale together.

$$F_{U,amp} = (\kappa_F m_r |\omega| / 2) g_{perp}$$

$$m_r = \rho \pi (r_o^2 - r_i^2) L$$

Therefore $F_{U,amp}$ is proportional to $\rho (r_o^2 - r_i^2) L |\omega|$

$$\Delta m_{amp} = F_{U,amp} / g_0$$

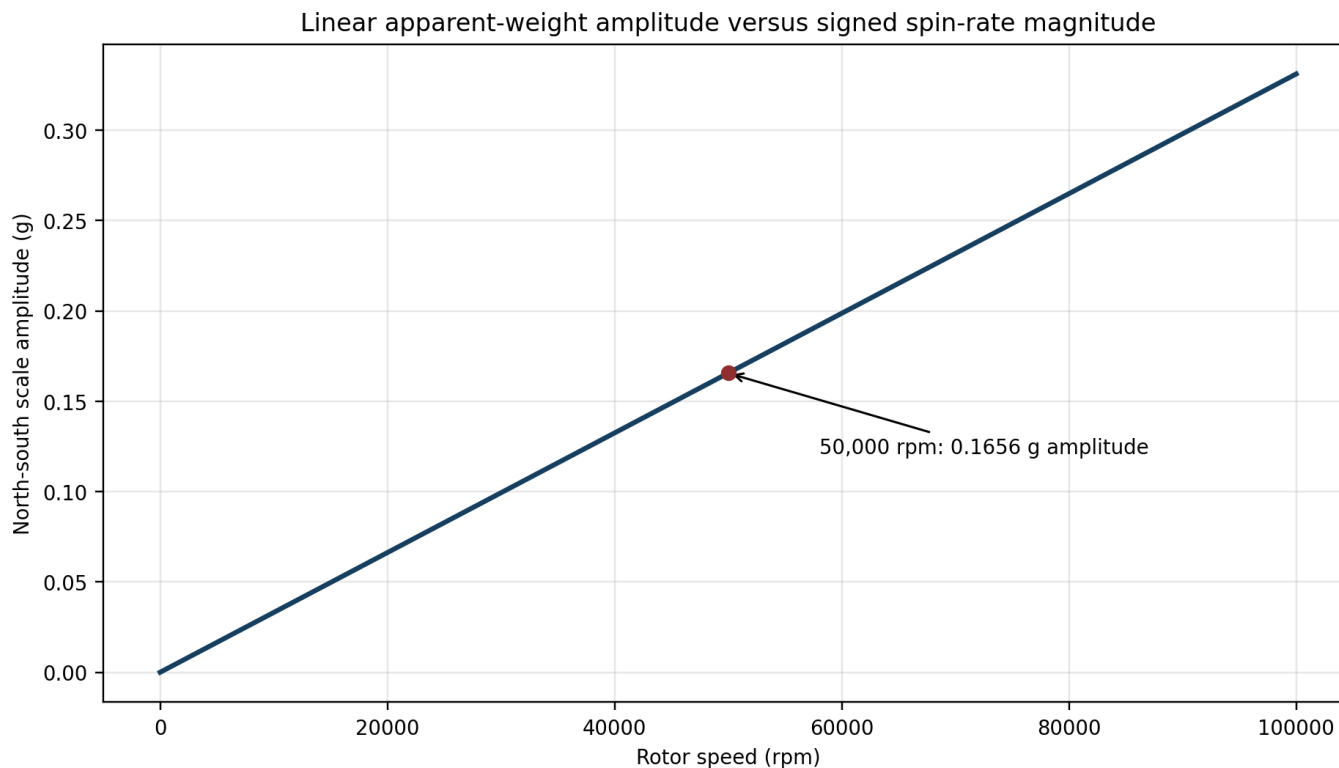


Figure 11. North-south apparent-mass amplitude versus rotor speed, derived from the complete source spin-rate ledger.

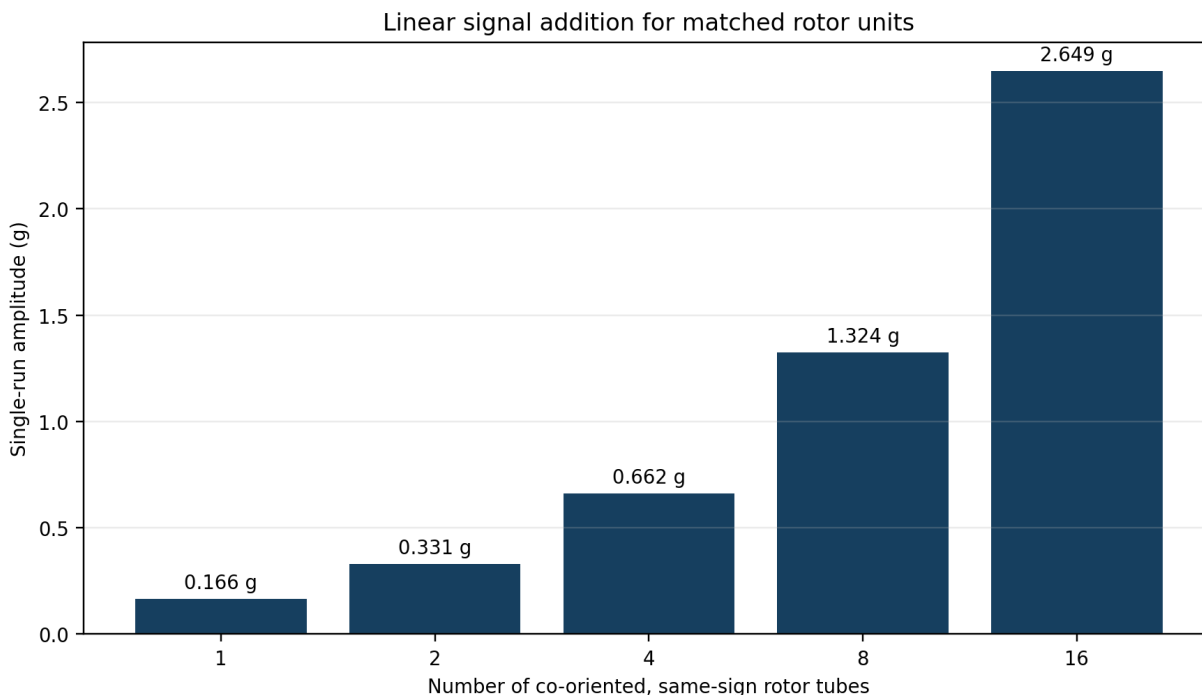


Figure 12. Linear addition for co-oriented rotor units with the same signed spin. Equal counter-rotating units cancel the ideal K-Field force.

Counter-rotation is commonly used to cancel conventional gyroscopic torque. In this force law it also cancels the desired signal when rotor axes and active masses are equal. Mechanical torque cancellation must therefore be achieved without canceling the signed K-Field sum, for example by structural reaction management, asymmetric active masses, or a reference rotor located off the primary measurement channel.

14. Assembly architecture

The minimum apparatus consists of a dynamically balanced tubular rotor, low-drift bearings, a motor or air turbine, an enclosure, a rigid north-south alignment frame, a scale with sub-0.01 g short-term resolution, independent speed measurement, temperature sensors, vibration sensing, and timestamped data acquisition referenced to local sidereal time. The predicted 0.1656 g single-run amplitude is large relative to modern laboratory mass resolution, but mechanical artifacts at 50,000 rpm can also be large. The architecture must therefore convert a high-speed rotating system into a quiet low-frequency scale measurement.

Mechanical arrangement

The rotor axis should be horizontal and level, aligned to true north-south, with the center of mass directly above the scale load center. The enclosure should be rigid enough that horizontal force does not create a changing moment about the scale platform. Symmetric feet or a single central load post reduce sensitivity to load-position drift. Flexible motor cables should be strain-relieved to the scale-mounted assembly so cable force does not bypass the scale.

Environmental arrangement

A closed enclosure suppresses aerodynamic thrust and room-air buoyancy changes. Vacuum operation is preferred for the final measurement because it removes windage, convective temperature gradients, and pressure-driven lift. Magnetic motor fields should be contained and the scale should be tested for magnetic susceptibility. The scale platform, rotor frame, and enclosure should reach thermal equilibrium before each block.

Metrology channels

Channel	Required measurement	Purpose
Scale	normal force or equivalent mass	primary K-Field projection
Tachometer	signed speed magnitude	normalize by omega
Temperature	rotor housing, scale, ambient	thermal regression and block matching
Vibration	vertical and horizontal acceleration	reject imbalance and resonance
Tilt	two-axis platform angle	correct gravity projection and cross-axis coupling
Motor electrical	voltage, current, phase	identify torque and magnetic state
Time	UTC and local sidereal time	phase-lock the predicted waveform

15. Spin-reversal experimental procedure

The experiment combines a time-domain sidereal orientation sweep with controlled spin reversal. A complete daily run maps the full projection law, while rapid matched-speed reversals provide the immediate opposite-force demonstration at any selected sidereal phase. Multiple days establish repeatability and sidereal phase coherence.

Step	Operation
1	Align the rotor axis horizontally to true north-south. Record azimuth and level uncertainty.
2	Place the complete self-contained assembly on the scale. Route no external mechanical support around the load cell.
3	Acquire a stopped baseline long enough to characterize drift, thermal settling, and environmental periodicity.
4	Spin to the target speed and hold speed constant. Record scale, tachometer, temperature, vibration, tilt, motor state, UTC, and local sidereal time.
5	Repeat at several speed magnitudes. The sidereal amplitude must scale linearly with $ \omega $.
6	Near the predicted optimum phase, acquire matched steady $+\omega$ and $-\omega$ blocks. The scale separation should approach 0.3311 g for the modeled tube at 50,000 rpm. Repeat the reversal at other sidereal phases to map the $2 F_U(t) $ envelope.
7	Rotate the entire apparatus by 90 degrees to east-west. The harmonic must change according to latitude and a spin-odd DC term may appear.
8	Set the rotor vertical. The direct ideal scale modulation must vanish.
9	Repeat with a nonspinning dummy rotor or locked rotor at the same motor thermal load where feasible.
10	Fit all runs jointly to the predicted vector model rather than accepting a generic sinusoid with unconstrained phase.

The north-south, east-west, and vertical orientations form a compact geometric triad. The north-south run maximizes the predicted daily harmonic, the east-west run tests the latitude-dependent transformation, and the vertical run measures residual mechanical coupling.

16. Conventional disturbance rejection

The expected signal is low frequency, spin odd, speed linear, axis dependent, and sidereal phase locked. Disturbances are rejected by testing all of these properties simultaneously.

Disturbance	Typical channel	Discriminating control
Rotor imbalance	vibration and load-cell oscillation at spin frequency	spectral monitoring; balance; low-pass average; speed dependence not purely signed
Bearing heating	slow scale drift	temperature matching; reversal; stopped thermal controls
Aerodynamic thrust	pressure and airflow dependent force	sealed enclosure; vacuum; reverse motor without rotor where possible
Motor magnetic force	interaction with scale or surroundings	magnetic shielding; motor-current regression; apparatus rotation controls
Cable force	external mechanical bypass	all cables strain-relieved to scale-mounted assembly; battery operation
Gyroscopic torque	load redistribution during precession or tilt	rigid axis; tilt monitoring; centered load path
Scale creep	slow monotonic or history-dependent drift	ABBA reversal blocks; independent reference mass; even/odd decomposition
Earth tides	common gravitational variation	spin-even cancellation; comparison to stopped reference
Buoyancy	air density and thermal gradients	sealed or vacuum enclosure; pressure and temperature recording
Floor vibration	common mechanical noise	isolation platform; reference load cell; coherence rejection

No single control is decisive by itself. The combined signature is decisive: the measured amplitude must be odd under spin reversal, linear in signed omega, maximal for north-south, suppressed for vertical, transformed correctly for east-west, and fixed in local sidereal phase across dates.

17. Data reduction and acceptance signatures

For each sample, compute local sidereal angle theta and the predicted basis vectors E, N, and U. The preferred analysis fits the full vector equation using measured omega rather than fitting only an unconstrained sinusoid.

```
g(t) = G_phase V_E(t) [fixed source value for single-day reference]
F_U,pred(t) = (kappa_F m_r omega(t)/2) [U(t) x n(t)] dot g(t)
m_scale,pred(t) = M_0 - F_U,pred(t)/g0
residual(t) = m_scale,measured(t) - m_scale,pred(t)
```

A minimal fixed-vector regression can use four components: constant offset, linear drift, predicted positive-spin template, and measured thermal terms. Opposite-spin data should share the same nuisance coefficients while the predicted template changes sign.

Acceptance signatures

Signature	Predicted result
Period	one sidereal day, not one solar day
North-south phase	apparent-mass maximum near LST 6.619026 h for the adopted +N,+omega convention
Spin reversal	180 degree sign reversal at matched speed
Speed scaling	amplitude proportional to signed omega
Axis scaling	north-south maximum; vertical zero; east-west follows sin(phi) harmonic law
Rotor count	linear sum for co-oriented same-sign rotors
Field-vector fit	phase and amplitude determined by g_x and g_y, not free parameters
Residuals	no remaining coherent spin-odd sidereal structure after full model fit

The primary result should be reported in force units and equivalent grams. Force units preserve direct comparison across scales and assembly masses; equivalent grams provide immediate metrological interpretation.

18. Orbital and seasonal extensions

The daily derivation holds g fixed over one run. Over longer intervals, Earth orbital velocity, solar-system barycentric motion, and any time-dependent response transform alter $V_E(t)$. The general equation remains unchanged; only the input vector becomes time dependent.

$$g(t) = G_{\text{phase}} V_E(t)$$

$$F_U(t) = (\kappa_F m_r \omega(t)/2) [U(t) \times n(t)] \cdot g(t)$$

A time-dependent g introduces annual amplitude modulation, phase drift, and sidebands around the sidereal line. These effects must be computed from an explicit ephemeris ledger before numerical values are assigned. The present source supplies one fixed V_E vector, so this report gives exact daily predictions for that source state and the reconstruction procedure for later ephemeris-grade extension.

The fixed-laboratory scale is therefore both a direct detector and a vector mapper. Daily rotation probes the equatorial response components. Seasonal variation changes the input velocity and supplies additional independent projections. Multiple laboratories at different longitudes observe the same local-sidereal phase after coordinate transformation, while different latitudes provide cross-checks for nonoptimal axis orientations.

19. Interpretive findings

Finding	Meaning
A fixed scale is sufficient	The apparatus need not be translated or mechanically reoriented during the run. Earth rotation performs the directional sweep relative to the ICRS-locked response.
The scale measures a projection	The measured quantity is not total force magnitude. It is $-F_K \cdot U$, the change in support force along local gravity.
North-south is the optimum passive axis	A horizontal north-south rotor samples the complete equatorial magnitude of g and has no spin-odd DC offset.
The signal is naturally modulated	The expected response is a first sidereal harmonic with phase fixed by RA_g and sign fixed by spin.
Reversal provides an internal null	The desired channel reverses exactly, while most weight and drift terms remain unchanged.
Magnitude is directly measurable	The source parameters predict 0.1656 g mean-to-peak for one 22.619 g tube at 50,000 rpm.
Geometry supplies multiple controls	Vertical, east-west, and north-south orientations provide distinct predicted outcomes from the same force law.
Scaling is explicit	Signal grows linearly with active rotor mass, tube length for fixed section, spin rate, and number of same-sign units.
The experiment maps field direction	Sidereal phase identifies the equatorial direction of g , while amplitude identifies its equatorial magnitude after coupling and rotor calibration.

The central conceptual transition is from free-space force magnitude to laboratory projection metrology. The rotor does not need to lift freely for its force to be observable. Constraint transfers the force into the scale normal reaction, and Earth rotation converts the fixed inertial direction into a periodic local-weight signal.

20. Conclusions

The supplied integrated tubular-rotor force law leads directly to an apparent-weight experiment at a fixed Earth laboratory. The complete assembly is placed on a scale, the rotor axis is held horizontal and true north-south, and the rotor is operated at controlled signed speed. Earth translation supplies the source velocity in $g = G_{\text{phase}} V_E$. Earth rotation sweeps the local basis through that ICRS-locked response and modulates the vertical force relative to local gravity.

The exact scale equation is $W_{\text{scale}} = M_{\text{assembly}} g_0 - F_K \cdot U$. For the north-south axis, $F_U = C[g_x \sin(\theta) - g_y \cos(\theta)]$, with amplitude $C \sqrt{g_x^2 + g_y^2}$. This orientation is globally maximal for passive daily modulation and is independent of laboratory latitude. A vertical rotor produces no direct ideal signal. An east-west rotor produces a latitude-reduced harmonic and a spin-odd constant offset.

For the modeled 10 mm outside-diameter, 8 mm inside-diameter, 100 mm long 316L working-density tube at 50,000 rpm, the predicted vertical-force amplitude is $1.623612850722e-03$ N. The corresponding scale amplitude is 0.165562 g and the single-run peak-to-peak range is 0.331125 g. Spin reversal doubles the full difference signal to 0.331125 g amplitude and 0.662250 g peak-to-peak.

The complete experimental signature is overdetermined: immediate equal-and-opposite steady readings under spin reversal, a midpoint equal to the stopped baseline, sidereal modulation of the instantaneous reversal separation, fixed local-sidereal phase, odd spin parity, linear signed-speed scaling, north-south maximum, vertical null, latitude-dependent east-west transformation, and linear same-sign rotor addition. Together these properties convert the fixed-laboratory scale into a direct, quantitative K-Field force and direction instrument without requiring a 12-hour wait for the first full-envelope demonstration.

Appendix A. Complete source numerical ledger

The following tables contain the complete numerical inputs used by this report. Values are preserved from the supplied source file except for explicitly labeled derived quantities.

Response matrix

Row	G_phase column 1	G_phase column 2	G_phase column 3
1	2.976410637800	1.243150779500	-2.080360035400
2	1.243150779500	2.592228663500	-2.317883993700
3	-2.080360035400	-2.317883993700	6.773570913300

Scalar and vector inputs

Field	Value
source schema	K-Field.tubular_rotor.free_space_forces.v1
source epoch	1784904809
source generated UTC	2026-07-24T14:53:29Z
source title	K-Field Tubular Rotor Free-Space Forces
source classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
V_E ICRS m/s	[-359003.19906581304, 76679.88603400199, -44701.060100687995]
G_phase V_E	[-880221.9816350914, -143911.43656159946, 266335.0269486193]
G_phase V_E norm	930825.217270434834063
kappa_F recommended	3.07405089970203453e-11
kappa_F interval	[3.04273994475465e-11, 3.10536185464942e-11]
tube density kg/m^3	8000.000000000000
tube OD m	0.010000000000
tube ID m	0.008000000000
tube length m	0.100000000000
tube mass kg	0.02261946710584652
rpm	50000.000000000000
omega rad/s	5235.98775598298834666
source force N	1.69445558791510081e-03
source acceleration m/s^2	7.49113840739921849e-02

Appendix B. All 27 Cell3 address nodes

Integer address	ICRS x native	ICRS y native	ICRS z native	Class	Index sum
(-1, -1, -1)	-0.010974902499961	0.007385983127423	0.039661911011939	corner	-3
(-1, -1, 0)	-0.011290092242635	-0.004742877881141	0.018199733726392	edge center	-2
(-1, -1, 1)	-0.011605281985309	-0.016871738889705	-0.003262443559155	corner	-1
(-1, 0, -1)	-0.009875529515800	0.013837068256769	0.022628660243911	edge center	-2
(-1, 0, 0)	-0.010190719258473	0.001708207248205	0.001166482958364	face center	-1
(-1, 0, 1)	-0.010505909001147	-0.010420653760359	-0.020295694327182	edge center	0
(-1, 1, -1)	-0.008776156531638	0.020288153386115	0.005595409475884	corner	-1
(-1, 1, 0)	-0.009091346274312	0.008159292377552	-0.015866767809663	edge center	0
(-1, 1, 1)	-0.009406536016985	-0.003969568631012	-0.037328945095210	corner	1
(0, -1, -1)	-0.000784183241488	0.005677757879218	0.038495428053575	edge center	-2
(0, -1, 0)	-0.001099372984162	-0.006451085129346	0.017033250768028	face center	-1
(0, -1, 1)	-0.001414562726835	-0.018579946137910	-0.004428926517519	edge center	0
(0, 0, -1)	0.000315189742674	0.012128861008564	0.021462177285547	face center	-1
(0, 0, 0)	0.000000000000000	0.000000000000000	0.000000000000000	body center	0
(0, 0, 1)	-0.000315189742674	-0.012128861008564	-0.021462177285547	face center	1
(0, 1, -1)	0.001414562726835	0.018579946137910	0.004428926517519	edge center	0
(0, 1, 0)	0.001099372984162	0.006451085129346	-0.017033250768028	face center	1
(0, 1, 1)	0.000784183241488	-0.005677757879218	-0.038495428053575	edge center	2
(1, -1, -1)	0.009406536016985	0.003969568631012	0.037328945095210	corner	-1
(1, -1, 0)	0.009091346274312	-0.008159292377552	0.015866767809663	edge center	0
(1, -1, 1)	0.008776156531638	-0.020288153386115	-0.005595409475884	corner	1
(1, 0, -1)	0.010505909001147	0.010420653760359	0.020295694327182	edge center	0
(1, 0, 0)	0.010190719258473	-0.001708207248205	-0.001166482958364	face center	1
(1, 0, 1)	0.009875529515800	-0.013837068256769	-0.022628660243911	edge center	2
(1, 1, -1)	0.011605281985309	0.016871738889705	0.003262443559155	corner	1
(1, 1, 0)	0.011290092242635	0.004742877881141	-0.018199733726392	edge center	2
(1, 1, 1)	0.010974902499961	-0.007385983127423	-0.039661911011939	corner	3

Authoritative Cell3 base cell: six faces, 12 edges, all 27 nodes

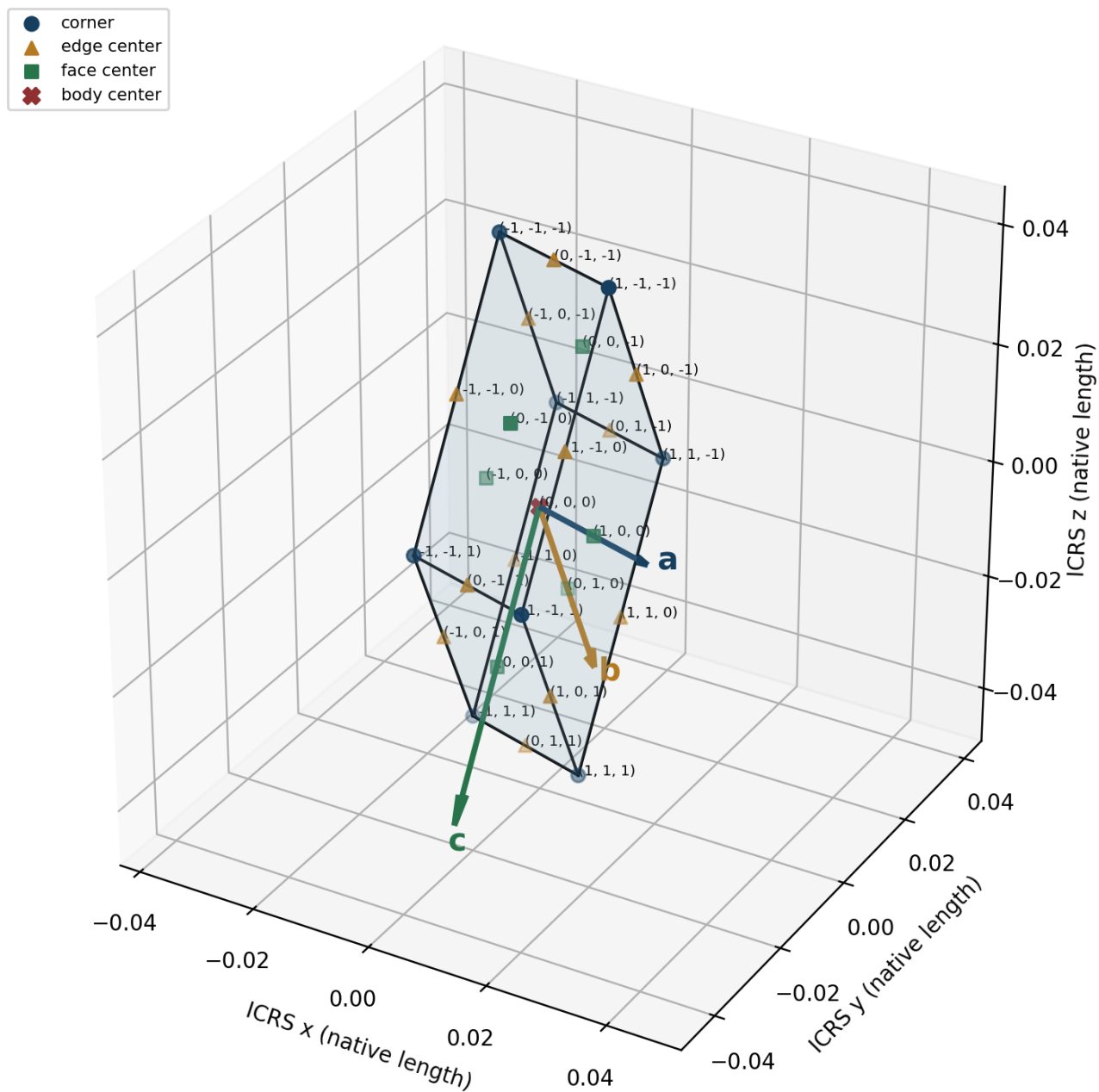


Figure B1. All node coordinates are rendered directly from the table above and the source basis matrix.

Appendix C. Sidereal scale sweep

The table lists one-hour samples for the positive north-south spin convention, its reversed state, and the direct opposite-spin separation. The machine-readable companion contains quarter-hour samples, the full source spin-rate and length-derived reversal sweeps, and the exact derivation fields.

LST h	Theta deg	F_up N	Scale shift g	Reversed shift g	Full difference g
0	0.0	2.619734976259e-04	-0.026713862	0.026713862	-0.053427725
1	15.0	-1.616687458290e-04	0.016485624	-0.016485624	0.032971248
2	30.0	-5.742935314259e-04	0.058561643	-0.058561643	0.117123285
3	45.0	-9.477811619210e-04	0.096646782	-0.096646782	0.193293563
4	60.0	-1.256679072514e-03	0.128145602	-0.128145602	0.256291205
5	75.0	-1.479936381075e-03	0.150911512	-0.150911512	0.301823024
6	90.0	-1.602338470976e-03	0.163393052	-0.163393052	0.326786104
7	105.0	-1.615543842070e-03	0.164739625	-0.164739625	0.329479250
8	120.0	-1.518652570140e-03	0.154859465	-0.154859465	0.309718930
9	135.0	-1.318267635246e-03	0.134425888	-0.134425888	0.268851776
10	150.0	-1.028044939550e-03	0.104831409	-0.104831409	0.209662819
11	165.0	-6.677626801492e-04	0.068092843	-0.068092843	0.136185686
12	180.0	-2.619734976259e-04	0.026713862	-0.026713862	0.053427725
13	195.0	1.616687458290e-04	-0.016485624	0.016485624	-0.032971248
14	210.0	5.742935314259e-04	-0.058561643	0.058561643	-0.117123285
15	225.0	9.477811619210e-04	-0.096646782	0.096646782	-0.193293563
16	240.0	1.256679072514e-03	-0.128145602	0.128145602	-0.256291205
17	255.0	1.479936381075e-03	-0.150911512	0.150911512	-0.301823024
18	270.0	1.602338470976e-03	-0.163393052	0.163393052	-0.326786104
19	285.0	1.615543842070e-03	-0.164739625	0.164739625	-0.329479250
20	300.0	1.518652570140e-03	-0.154859465	0.154859465	-0.309718930
21	315.0	1.318267635246e-03	-0.134425888	0.134425888	-0.268851776
22	330.0	1.028044939550e-03	-0.104831409	0.104831409	-0.209662819
23	345.0	6.677626801492e-04	-0.068092843	0.068092843	-0.136185686
24	360.0	2.619734976259e-04	-0.026713862	0.026713862	-0.053427725

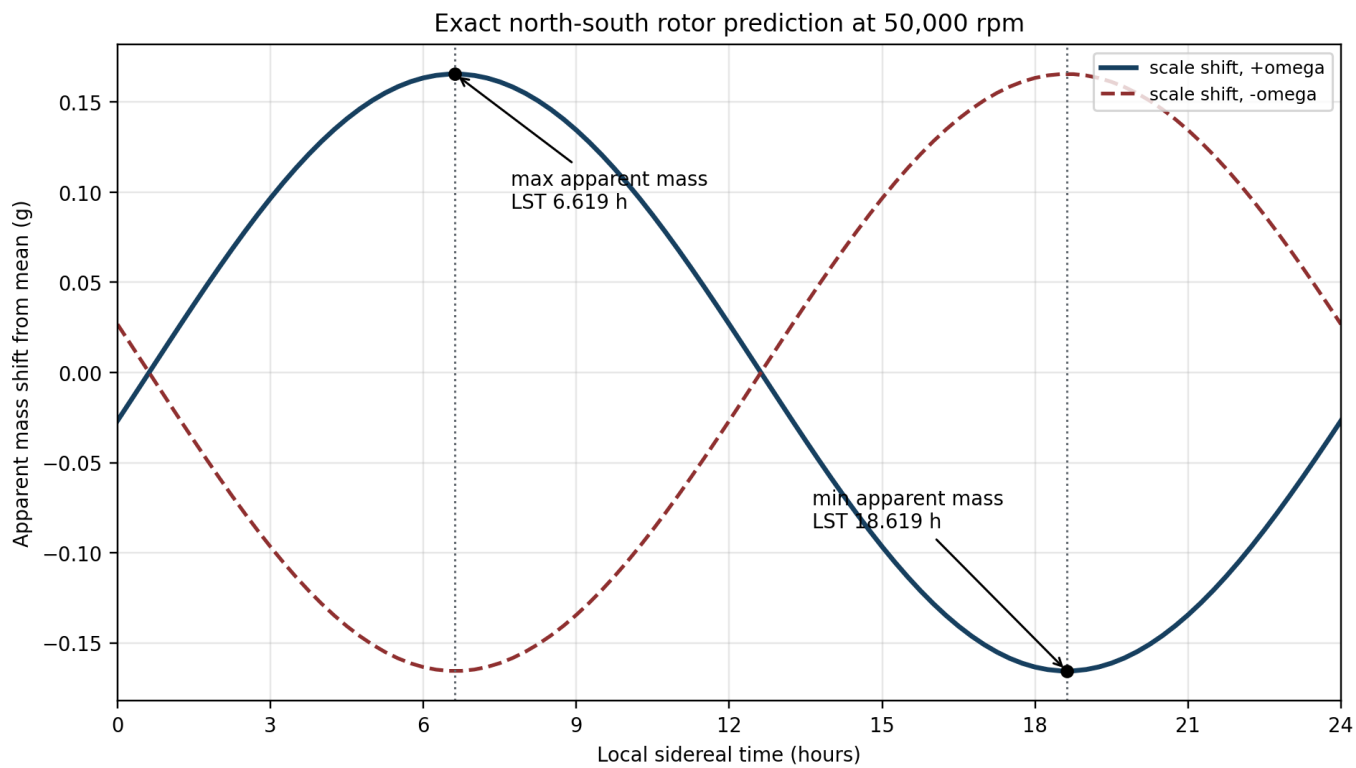


Figure C1. Continuous interpolation of the exact sidereal sweep.

Appendix D. Compact reconstruction equations

Step	Equation
1. Source response	$g = G_{\text{phase}} V_E$
2. Tube mass	$m_r = \rho \pi (r_o^2 - r_i^2) L$
3. Signed angular speed	$\omega = \text{RPM} (2 \pi / 60)$
4. Integrated tube force	$F_K = (\kappa_F m_r \omega / 2) (n \times g)$
5. Local basis	$U = (\cos \phi \cos \theta, \cos \phi \sin \theta, \sin \phi); E = (-\sin \theta, \cos \theta, 0); N = (-\sin \phi \cos \theta, -\sin \phi \sin \theta, \cos \phi)$
6. Scale force	$W_{\text{scale}} = M_{\text{assembly}} g_0 - F_K \cdot U$
7. General fixed-axis vertical force	$F_U = (\kappa_F m_r \omega / 2) (U \times n) \cdot g$
8. Horizontal azimuth axis	$n = \cos \alpha N + \sin \alpha E$
9. Harmonic amplitude	$A(\phi, \alpha) = C_{\text{g_perp}} \sqrt{\cos^2 \alpha + \sin^2 \alpha \sin^2 \phi}$
10. North-south waveform	$F_U = C[g_x \sin \theta - g_y \cos \theta] = C_{\text{g_perp}} \sin(\theta - \text{RA}_g)$
11. Apparent mass	$\Delta m_{\text{scale}} = -F_U / g_0$
12. Spin-odd channel	$W_{\text{odd}} = [W(+\omega) - W(-\omega)] / 2$
13. Full reversal difference	$W(+\omega) - W(-\omega) = -2F_U(+\omega)$
14. Same-sign rotor sum	$F_{\text{total}} = \sum_i (\kappa_F m_i \omega_i / 2) (n_i \times g)$

Reconstruction procedure

Read the source basis and field-response matrix. Compute g from the declared Earth velocity. Compute tube mass and signed angular speed. Transform the fixed laboratory axis into the ICRS local basis for each sample time. Evaluate the scalar triple product $(U \times n) \cdot g$. Convert vertical force to equivalent mass by division by g_0 . For reversal data, preserve the same geometry and change only the sign of ω .

Appendix E. Archival record

Field	Record
Document title	K-Field Apparent-Weight Modulation Spin-Reversal Experiment
Principal Investigator	P. Dwayne Esterline
Education	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation UTC	2026-07-24T17:20:57Z
Generation epoch	1784913657
Output PDF	K-Field_apparent_weight_modulation_spin_reversal_experiment_1784913657.pdf
Machine-readable companion	K-Field_apparent_weight_modulation_spin_reversal_experiment_1784913657.json
Source JSON	K-Field_tubular_rotor_free_space_forces_1784904809(1).json
Historical format references	K-Field_tubular_rotor_free_space_forces_1784904809.pdf; K-Field_dual_axis_kinetic_mass_CMB_traversal_geometry_1784745791.pdf
Coordinate frame	ICRS/J2000 Cartesian; local east-north-up projection
Mechanical units	SI; equivalent scale mass also reported in grams
Cell3 units	native source units
Report generation	Python ReportLab; built-in core fonts only
Figure generation	Python numerical computation and raster rendering from exact report ledger
Page size	US Letter
Margins	0.68 inch on all sides
Table widths	dynamically normalized to available page width
Cell wrapping	every table text cell is a ReportLab Paragraph
Long-table headers	repeatRows=1

P. Dwayne Esterline

Krytronx

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